

Kasa bez zarządu

Savings, 0% loans, direct debits and standing orders with **no board, no treasurer and no bank**. Every rule is a Solana program with no admin key.

0

people who approve a loan

100%

of every loan covered by locked savings

0 clicks

installments leave the wallet by themselves, sent by a bot with no rights

THE PROBLEM

Pooling money only works through intermediaries you must trust

Who we build for: small groups in Poland that already pool money. Workplace *kasy zapomogowo-pożyczkowe* (KZP) in schools, offices and factories, teams with a "kasa koleżeńska", friends and families lending to each other.

Members save every month and borrow at 0%, with co-members vouching. It works only through:

- **the treasurer**, who holds the money and the books,
- **the board**, which decides who gets a loan,
- **the employer's payroll**, which deducts installments.

Without such a group, the same services come from a **bank**: credit with interest, standing orders, direct debits, a BIK score.

When that trust fails

A PKZP treasurer in Jawor county took **61,000 zł** of colleagues' savings and lost it on a crypto exchange.

| Puls Legnicy, March 2024

A PKZP cashier hid **261,510 zł** of withdrawals with falsified cash reports for **ten years**.

| Court of Appeal in Katowice, II AKa 98/15

Informal groups have it worse: "the kasa" is simply someone's private bank account.

THE REDESIGN

Every intermediary, and every bank service, becomes a rule

SERVICE	INTERMEDIARY TODAY	KASA BEZ ZARZĄDU
Keeping savings	Treasurer, bank	Vault owned by the program; only the rules move tokens out
Credit	Board, bank scoring	Nobody approves: pays out when own + guarantors' locked savings cover 100%, at 0%
Direct debit of installments	Employer's payroll, bank	Revocable SPL mandate to the member's own PDA; from the due date anyone pulls <i>only what is due</i>
Debt collection	Board, collection agency	After grace anyone collects from collateral: borrower first, then guarantors
Standing order to save	Bank	Standing contribution, pulled once per period under the same mandate
Statement & credit history	Bank, BIK	Kasa history from the chain with the balance after every move; a guarantor sees the borrower's record across all kasas
Changing the rules	General meeting, board	No instruction changes them; no admin key

HOW IT WORKS

Save, borrow with guarantors, and let the installments go by themselves

1 · SAVE

Into the vault

Anyone joins. Optional standing contribution, pulled every period.

2 · BORROW

Up to 2× savings

Own free savings lock first; co-members pledge the rest.

3 · PAY OUT + MANDATE

100% covered → paid

One transaction: loan to the wallet and an SPL allowance for the installments.

4 · REPAY

On the due date

The bot pulls what is due from the wallet; collateral unlocks, guarantors first.

Installment 1: nobody clicks

Bartek borrowed 1000 zł in 4 installments (own 500, Anna 300, Celina 200). On the due date the bot pulls **250 zł from his wallet**. He signs nothing. Anna and Celina get 250 zł unlocked, 3:2.

He revokes and stops paying

Installment 2: the bot's pull is refused (NoMandate). After grace the bot collects 250 zł from **Bartek's locked savings**; later installments take his last 250 zł, then Anna and Celina pro rata.

The fund ends whole

Bartek kept the cash, lost his savings. Anna –150 zł, Celina –100 zł: the risk they chose. **Fund: 0 zł lost.** Everyone withdraws the rest; vault ends at 0. All on devnet.

The installment leaves the wallet on the due date: no payroll, no bank, no click

Bartek pożycza 1 000 zł (Twoja)Splacana
4 raty po ok. 250 zł · co 1 min · [konto pożyczki](#)

Zablokowane oszczędności na resztę do spłaty: 750 zł z 750 zł

■ Bartek (swoje): 500 zł ■ Anna poręcza: 150 zł ■ Celina poręcza: 100 zł

Spłacone: **250 zł** Zostało: **750 zł** Spłacone automatycznie: **250 zł**

↻ **Automatyczna spłata włączona.** W dniu terminu rata sama zejdzie z Twojego portfela – bez banku i bez klikania. Program weźmie tylko ratę, nic więcej (limit: 1 000 zł, w portfelu: 6 750 zł).

Rata 1: 250 zł
spłacona

Rata 2: 250 zł
za 47 s · zapłaci się sama

Rata 3: 250 zł
za 1 min 47 s · zapłaci się sama

Rata 4: 250 zł
za 2 min 47 s · zapłaci się sama

🕒 Jeśli rata 2 nie wpłynie, za 1 min 2 s zostanie pokryta z zablokowanych oszczędności.

Spłacić (zł)

Spłaci ratę

Wyłącz automatyczną spłatę

```
// instructions/autopay.rs: no signer required
pub fn handle_pull_installment(ctx) {
    let due = loan.overdue(now, period, 0)?;
    require!(due > 0, InstallmentNotDue);
    // allowance the borrower gave to
    // THEIR member PDA, capped by balance
    let amount = due.min(mandate_available(
        &from, &borrower_member.key()));
    require!(amount > 0, NoMandate);
    pay_in_by_mandate(.., amount)?; // PDA signs
    book_repayment(..)?; // guarantors first
}
// from.owner = loan.borrower, or rejected
```

The SPL Token program enforces the allowance; ours enforces the amount and the date. The bot only pays the fee; if the mandate is gone, collateral pays after grace.

Two invariants, tested after every step

1. Every loan is fully covered

Pay-out requires locked savings = loan amount.
Repayments (by hand or by mandate) release collateral;
collection consumes it. Collateral always equals what is still owed.

2. The vault holds every free saving

$$\text{vault} = \Sigma \text{savings} - \Sigma \text{lent}$$

So anyone can withdraw what is theirs at any moment. The tests end with a bank run: everyone withdraws everything, vault at 0.

The mandate is a convenience, not the guarantee

The borrower can revoke it any time. The fund never depended on it: collateral covers what the wallet doesn't.

The only risk is a chosen one

Only a guarantor who vouched for someone who didn't pay can lose, at most the pledge. Now they can read the borrower's on-chain record first.

QUESTIONS YOU WILL ASK

Someone disappears? Who can do what? Can we change it?

WHO GOES SILENT	WHAT HAPPENS
Borrower	Revoked mandate or empty wallet: collateral pays after grace, then guarantors
Guarantor	Nothing to do: the pledge unlocks as the loan is repaid
The bot	Nothing breaks: no rights; anyone presses the same buttons or runs it
Founder, us, the site	Nothing: no rights; program + IDL on-chain

ACTION	WHO
Join, deposit, withdraw free savings	Anyone, for themselves
Give / revoke a mandate	Wallet owner (SPL approve / revoke)
Pull what is due under a mandate	Anyone: only due, only that wallet, within the allowance
Repay, collect overdue	Anyone
Change rules, move others' savings	No instruction allows it, for anyone

Honest caveat: on devnet our deploy key can still upgrade the program code. Kept for bug fixes during the hackathon; next: a members' multisig, or `--final`.

Every claim is backed by a test or a transaction

20/20

Rust tests, LiteSVM + real SPL Token: 8 new for the mandate (never more than due, never from another wallet, revoke → collateral)

14/14

Web tests: the app's math matches the program; credit history; the kasa history follows the vault

E2E

Devnet run: bot pulls a contribution and an installment with nobody signing, revoke, collection, bank run → vault 0

UI

Playwright drives 3 wallets through the deployed app while the bot runs on Railway

Historia kasy

Każdy grosz, który wszedł do kasy albo z niej wyszedł. Lista pochodzi prosto z łańcucha: nikt nie może jej poprawić ani niczego z niej usunąć.

Ruchy pieniędzy

Tylko wpływy

Wszystko

Wpłynęło: **3 050 zł** Wypłynęło: **1 000 zł** Jest w kasie: **2 050 zł** ✓ zgadza się z historią

Kiedy	Co się stało	Kogo dotyczy	Kwota	W kasie potem	Kto wysłał	Dowód
00:19:18	Niezapłacona rata pokryta z zablokowanych oszczędności pieniądze nie wychodzą z kasy	Bartek	250 zł	2 050 zł	🔗 automat	3fFaGhCVL4... 🔗
00:18:30	Składka pobrana automatycznie z portfela	Anna	+100 zł	2 050 zł	🔗 automat	4RcxfevFYA... 🔗
00:18:02	Rata pobrana automatycznie z portfela	Bartek	+250 zł	1 950 zł	🔗 automat	BA5E6Kj9Dt... 🔗
00:17:31	Składka pobrana automatycznie z portfela	Anna	+100 zł	1 700 zł	🔗 automat	DpkpzWmiD1... 🔗
00:16:58	Wypłata pożyczki	Bartek	-1 000 zł	1 600 zł	Bartek	2XnA7rfahG... 🔗
00:16:53	Wpłata	Celina	+1 000 zł	2 600 zł	Celina	524miAUWdz... 🔗
00:16:41	Wpłata	Bartek	+500 zł	1 600 zł	Bartek	2UsbFrtwgb... 🔗
00:16:30	Składka pobrana automatycznie z portfela	Anna	+100 zł	1 100 zł	🔗 automat	4JFpEpVbMe... 🔗
00:16:23	Wpłata	Anna	+1 000 zł	1 000 zł	Anna	5kC6qdMQ1K... 🔗

Why blockchain, and why not just "send USDC"

Why not a database?

Whoever runs it is the intermediary: they can edit a balance, hold back a withdrawal or "borrow" from the pot, which is how the cases on slide 2 happened. Here the balance is a program-owned account and the rules are public code no instruction can bypass.

Why not just transfers?

Wallet-to-wallet payments without a bank already exist. What still needs one is credit, a direct debit someone honours, a standing order and a credit record. Those are what the program replaces; a direct debit needs no trusted bank when the token program caps it and our program sets the amount.

Why not an escrow marketplace?

For physical goods someone must judge delivery: an arbiter is still an intermediary. We chose a relationship where **the program sees every condition it enforces**: deposits, balances, the schedule, the clock, the allowance.

What it doesn't do yet, and what we'd build next

Limitations

- Test złoty from a devnet faucet; the program takes any SPL mint (production: a stablecoin)
- Members need a Solana wallet
- Programs don't run on a timer: a bot (no rights) sends the pulls; if it stops, anyone can
- One SPL delegate per token account: a mandate in a second kasa replaces the first
- Credit history is public by design; should be opt-in for real groups
- The 2021 Act still expects a board; program still upgradeable by our key; not audited

With another week

- PLN stablecoin, BLIK on-ramp, passkey wallets
- Guarantees across kasas: vouch from your savings in another group
- *Zapomogi*: grants released by member vote
- Reminders before a pull; one mandate shared by all kasas
- Upgrade key to a members' multisig, audit, mainnet

Try it: web-production-ad49f.up.railway.app · **Code:** github.com/jxroo/kasa-bez-zarzadu